Harshini Bandaru

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EDUCATION

University of Pittsburgh **Pittsburgh, PA**
Bachelor of Science in Computer Science; Minor in Information Science (3.9/4.0) *August 2021 - May 2024*
Relevant Coursework: Cloud Computing, Database Systems, Operating Systems, Algorithms & Data Structures

SKILLS

Languages: Python, Go, Java, C, C++, SQL, JavaScript/TypeScript
Tools/Frameworks: Flask, FastAPI, Node.js, React, Git, Docker
Cloud/Deployment: AWS, GCP, Kubernetes, Azure, CI/CD
AI Tools: Claude Code, Codex, Github Copilot, Cursor, MCP

WORK EXPERIENCE

Software Engineer II **Chicago, IL**
Deloitte *October 2024 - Present*

- Built and maintained **5 backend data pipelines** in **Python** and **SQL**, orchestrated through **Azure DevOps**, to sync and transform data feeding production analytics dashboards across **3 enterprise applications**.
- Owned **Python/PostgreSQL** backend services powering **KPI dashboards** for 200+ users, handling validation, schema changes, and query optimization that cut load times from 12s to 3s.
- Served as technical point of contact for a **15-engineer** offshore team, using design-phase context to guide **Azure VM** performance tuning that reduced runtime by ~30%.
- Architected a **Python/OpenAI API** backend service generating **50 user stories per run**, cutting a multi-day BA process to minutes, with retry logic, output validation, and logging for reliability.

Software Engineering Intern **Apex, NC**
Pentair *May 2023 - August 2023*

- Architected an AI chatbot using **React Native** frontend and **Python** backend, integrating **REST APIs**, **OpenAI APIs**, and **AWS Lambda** to enable scalable, serverless inference and modular system design.
- Integrated chatbot with IoT system components, improving request handling reliability and reducing latency by **~25%** under load, while validating performance using **200+ Postman test cases**, reducing integration defects by **~30%**.

Software Engineering Intern **Pittsburgh, PA**
Federated Hermes *January 2023 - April 2023*

- Streamlined certificate-expiration tracking by building Boomi **ETL** flows, **XML** data mappings, transformation logic, and automated validations, significantly improving data accuracy and reducing manual intervention by **~95%**.
- Reviewed code across environments using **Git** and Visual Studio, strengthening deployment reliability and quality.

Software Engineering Intern **Warrendale, PA**
Federated Hermes *June 2022 - August 2022*

- Redesigned data access and reporting by building **Power Apps** dashboards and **Power BI** reports on optimized **ETL pipelines**, increasing accessibility by **25%**, efficiency by **20%**, and cutting processing time by **50%**.
- Developed a **Java** project management tool using **Postgres** for task scheduling, meetings, and deadlines, with documentation.

CS Teaching Assistant **Pittsburgh, PA**
University of Pittsburgh *January 2022 - April 2022*

- Led weekly lectures for 20+ undergraduates on **OOP** and **data structures**, coaching students to improve problem-solving.

PROJECTS

ClosetLocal November 2026

- Architected a privacy-first, cross-platform wardrobe and laundry management system using **React Native, Expo, TypeScript, FastAPI, SQLite, and SQLAlchemy**, owning data modeling, API design, and deployment.
- Designed a local computer-vision pipeline using **CLIP and SegFormer** for garment classification, segmentation, and visual search, then integrated a recommendation engine using wardrobe availability, laundry state, and live weather.

HealthConnect *July 2025*

- Built a **full-stack** healthcare consulting platform using **React + Node.js/Express** with **REST APIs** and **Socket.io**, with role-based challenge workflows, anonymous consultant bidding, and real-time updates for client-consultant interactions.
- Implemented **Firebase Auth/Admin RBAC** for secure user management, and **Firestore-backed** real-time chat, enabling persistent, **low-latency** messaging with per-challenge data isolation and live bid tracking.

Multi-Process and Multi-Threaded Web Servers *October 2023*

- Developed two implementations of a web server with **multithreading** & **multiprocessing**.
- Built concurrent **C++** HTTP servers using fork() and pthreads()s, with POSIX semaphores/mutexes for request logging.
- Scaled system to efficiently handle up to **500** simultaneous requests while maintaining **low latency** and reliable performance.